

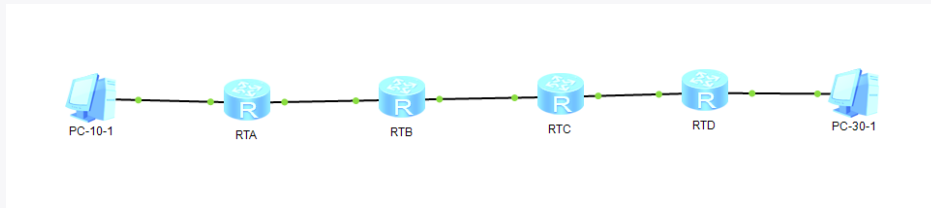
Lab-6 (1) Basic OSPF Configuration and Analysis(E)

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2026-05-22

Lab-6-1 BASIC MULTIPLE AREA OSPF CONFIGURATION

Q6-1.1. Paste the screenshot of the created topology.



Q6-1.2. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTA.

```
RTA
RTA  RTB  RTC  RTD
[RTA]
[RTA]
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[RTA]
[RTA]
[RTA]
[RTA]display ospf routing

      OSPF Process 1 with Router ID 10.1.1.1
      Routing Tables

Routing for Network
Destination      Cost  Type      NextHop      AdvRouter      Area
172.16.101.0/24  1     Transit   172.16.101.1  10.1.1.1       0.0.0.1
192.168.10.0/24  1     Stub      192.168.10.1  10.1.1.1       0.0.0.1
172.16.102.0/24  2     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
172.16.103.0/24  3     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
192.168.30.0/24  4     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
192.168.200.100/32 3     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1

Total Nets: 6
Intra Area: 2  Inter Area: 4  ASE: 0  NSSA: 0

[RTA]
```

```

RTA
RTA  RTB  RTC  RTD

Routing for Network
Destination      Cost  Type      NextHop      AdvRouter      Area
172.16.101.0/24  1     Transit   172.16.101.1  10.1.1.1       0.0.0.1
192.168.10.0/24  1     Stub      192.168.10.1  10.1.1.1       0.0.0.1
172.16.102.0/24  2     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
172.16.103.0/24  3     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
192.168.30.0/24  4     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1
192.168.200.100/32 3     Inter-area 172.16.101.2  10.2.2.2       0.0.0.1

Total Nets: 6
Intra Area: 2  Inter Area: 4  ASE: 0  NSSA: 0

[RTA]display ospf lsdb

      OSPF Process 1 with Router ID 10.1.1.1
      Link State Database

      Area: 0.0.0.1
Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
Router    10.2.2.2             10.2.2.2       359  36   800000004     1
Router    10.1.1.1             10.1.1.1       359  48   800000006     1
Network   172.16.101.2         10.2.2.2       359  32   800000002     0
Sum-Net   192.168.200.100     10.2.2.2       340  28   800000001     2
Sum-Net   172.16.103.0        10.2.2.2       349  28   800000001     2
Sum-Net   192.168.30.0        10.2.2.2       340  28   800000001     3
Sum-Net   172.16.102.0        10.2.2.2       397  28   800000001     1

[RTA]

```

```

RTA
RTA  RTB  RTC  RTD

      OSPF Process 1 with Router ID 10.1.1.1
      Link State Database

      Area: 0.0.0.1
Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
Router    10.2.2.2             10.2.2.2       359  36   800000004     1
Router    10.1.1.1             10.1.1.1       359  48   800000006     1
Network   172.16.101.2         10.2.2.2       359  32   800000002     0
Sum-Net   192.168.200.100     10.2.2.2       340  28   800000001     2
Sum-Net   172.16.103.0        10.2.2.2       349  28   800000001     2
Sum-Net   192.168.30.0        10.2.2.2       340  28   800000001     3
Sum-Net   172.16.102.0        10.2.2.2       397  28   800000001     1

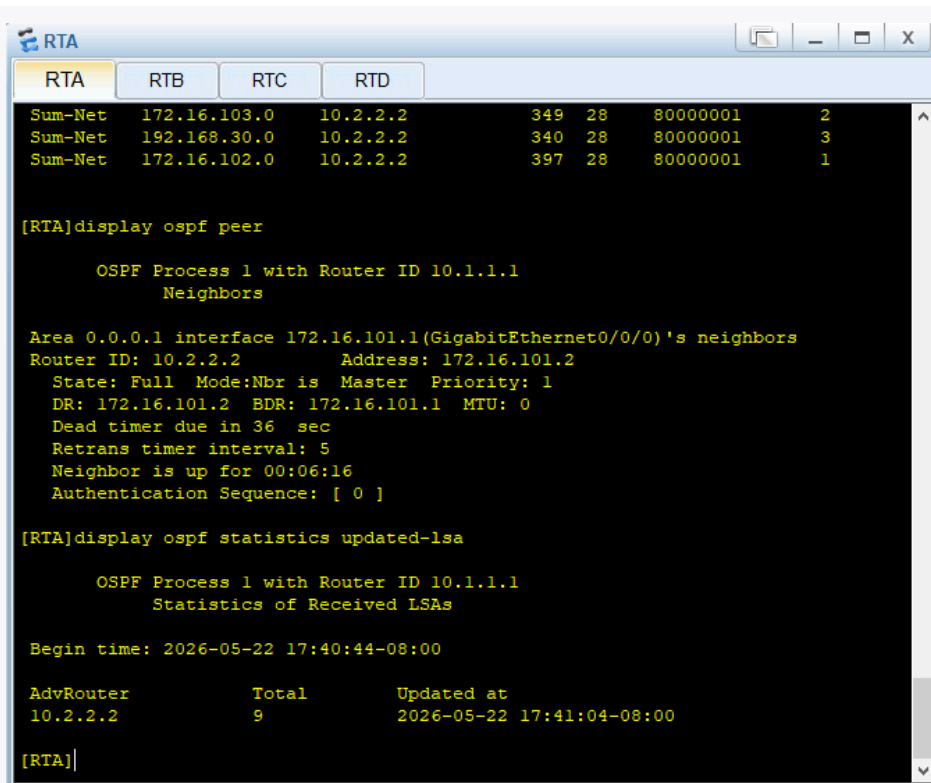
[RTA]display ospf peer

      OSPF Process 1 with Router ID 10.1.1.1
      Neighbors

Area 0.0.0.1 interface 172.16.101.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.2.2.2      Address: 172.16.101.2
State: Full  Mode:Nbr is Master  Priority: 1
DR: 172.16.101.2  BDR: 172.16.101.1  MTU: 0
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00:06:16
Authentication Sequence: [ 0 ]

[RTA]

```



```

RTA
RTA  RTB  RTC  RTD
Sum-Net  172.16.103.0  10.2.2.2  349 28  80000001  2
Sum-Net  192.168.30.0  10.2.2.2  340 28  80000001  3
Sum-Net  172.16.102.0  10.2.2.2  397 28  80000001  1

[RTA]display ospf peer

      OSPF Process 1 with Router ID 10.1.1.1
        Neighbors

Area 0.0.0.1 interface 172.16.101.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.2.2.2      Address: 172.16.101.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 172.16.101.2  BDR: 172.16.101.1  MTU: 0
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00:06:16
  Authentication Sequence: [ 0 ]

[RTA]display ospf statistics updated-lsa

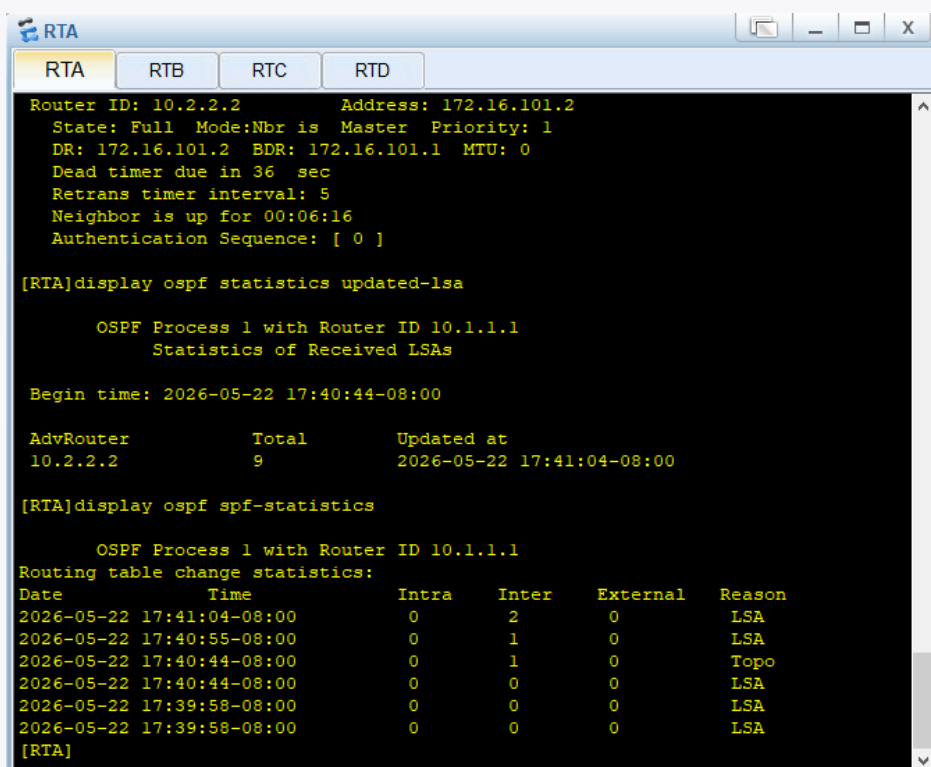
      OSPF Process 1 with Router ID 10.1.1.1
        Statistics of Received LSAs

Begin time: 2026-05-22 17:40:44-08:00

AdvRouter      Total      Updated at
10.2.2.2        9        2026-05-22 17:41:04-08:00

[RTA]

```



```

RTA
RTA  RTB  RTC  RTD

Router ID: 10.2.2.2      Address: 172.16.101.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 172.16.101.2  BDR: 172.16.101.1  MTU: 0
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00:06:16
  Authentication Sequence: [ 0 ]

[RTA]display ospf statistics updated-lsa

      OSPF Process 1 with Router ID 10.1.1.1
        Statistics of Received LSAs

Begin time: 2026-05-22 17:40:44-08:00

AdvRouter      Total      Updated at
10.2.2.2        9        2026-05-22 17:41:04-08:00

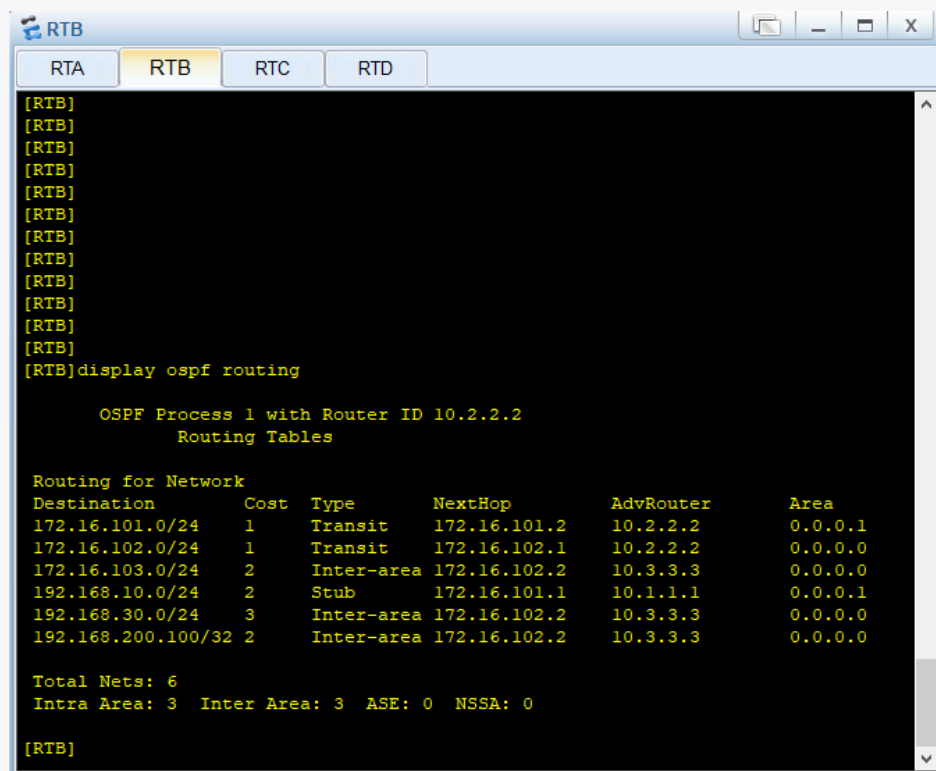
[RTA]display ospf spf-statistics

      OSPF Process 1 with Router ID 10.1.1.1
Routing table change statistics:
Date          Time          Intra   Inter   External  Reason
2026-05-22 17:41:04-08:00    0       2       0       LSA
2026-05-22 17:40:55-08:00    0       1       0       LSA
2026-05-22 17:40:44-08:00    0       1       0       Topo
2026-05-22 17:40:44-08:00    0       0       0       LSA
2026-05-22 17:39:58-08:00    0       0       0       LSA
2026-05-22 17:39:58-08:00    0       0       0       LSA

[RTA]

```

Q6-1.3. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTB.



```

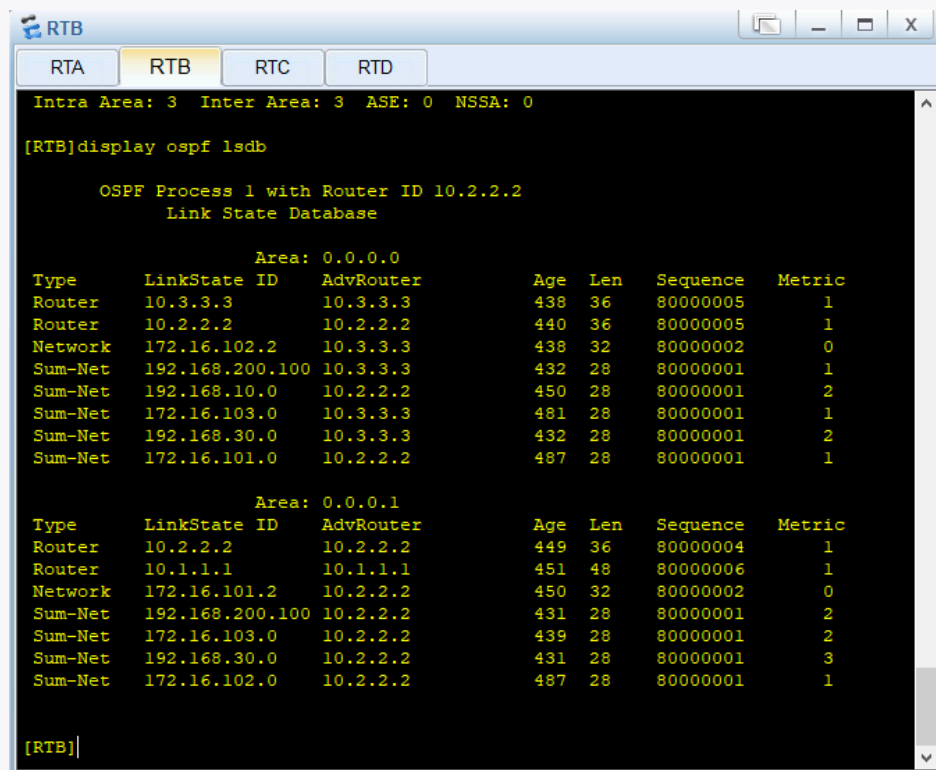
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]
[RTB]display ospf routing

      OSPF Process 1 with Router ID 10.2.2.2
        Routing Tables

Routing for Network
Destination      Cost  Type      NextHop      AdvRouter      Area
172.16.101.0/24  1     Transit   172.16.101.2  10.2.2.2        0.0.0.1
172.16.102.0/24  1     Transit   172.16.102.1  10.2.2.2        0.0.0.0
172.16.103.0/24  2     Inter-area 172.16.102.2  10.3.3.3        0.0.0.0
192.168.10.0/24  2     Stub      172.16.101.1  10.1.1.1        0.0.0.1
192.168.30.0/24  3     Inter-area 172.16.102.2  10.3.3.3        0.0.0.0
192.168.200.100/32 2     Inter-area 172.16.102.2  10.3.3.3        0.0.0.0

Total Nets: 6
Intra Area: 3  Inter Area: 3  ASE: 0  NSSA: 0

[RTB]
  
```



```

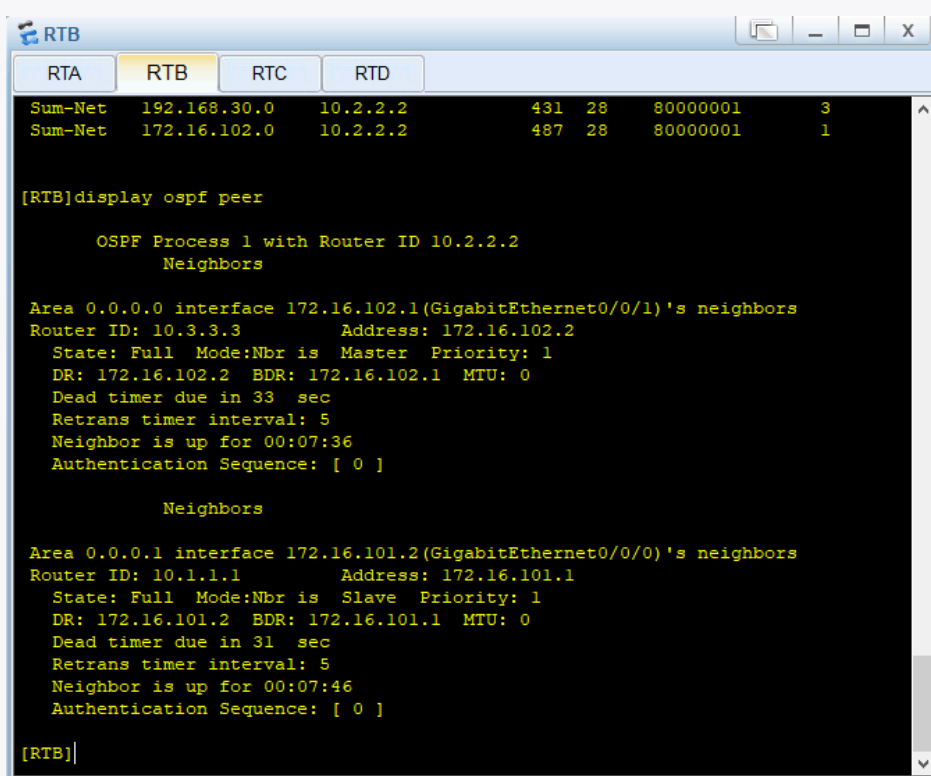
[RTB]display ospf lsdb

      OSPF Process 1 with Router ID 10.2.2.2
        Link State Database

                Area: 0.0.0.0
Type  LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router 10.3.3.3        10.3.3.3        438  36  80000005      1
Router 10.2.2.2        10.2.2.2        440  36  80000005      1
Network 172.16.102.2    10.3.3.3        438  32  80000002      0
Sum-Net 192.168.200.100 10.3.3.3        432  28  80000001      1
Sum-Net 192.168.10.0   10.2.2.2        450  28  80000001      2
Sum-Net 172.16.103.0   10.3.3.3        481  28  80000001      1
Sum-Net 192.168.30.0   10.3.3.3        432  28  80000001      2
Sum-Net 172.16.101.0   10.2.2.2        487  28  80000001      1

                Area: 0.0.0.1
Type  LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router 10.2.2.2        10.2.2.2        449  36  80000004      1
Router 10.1.1.1        10.1.1.1        451  48  80000006      1
Network 172.16.101.2    10.2.2.2        450  32  80000002      0
Sum-Net 192.168.200.100 10.2.2.2        431  28  80000001      2
Sum-Net 172.16.103.0   10.2.2.2        439  28  80000001      2
Sum-Net 192.168.30.0   10.2.2.2        431  28  80000001      3
Sum-Net 172.16.102.0   10.2.2.2        487  28  80000001      1

[RTB]
  
```



The screenshot shows the RTB terminal window with tabs for RTA, RTB, RTC, and RTD. The RTB tab is active. The terminal displays the output of the command `[RTB]display ospf peer`. It shows OSPF Process 1 with Router ID 10.2.2.2. The neighbors are listed for two areas: Area 0.0.0.0 and Area 0.0.0.1. For Area 0.0.0.0, the neighbor is 172.16.102.1 (GigabitEthernet0/0/1) with Router ID 10.3.3.3. For Area 0.0.0.1, the neighbor is 172.16.101.2 (GigabitEthernet0/0/0) with Router ID 10.1.1.1. The state for both is 'Full Mode:Nbr is Master Priority: 1'. The dead timer is 33 seconds for Area 0.0.0.0 and 31 seconds for Area 0.0.0.1. The neighbor is up for 00:07:36 in both areas. The authentication sequence is [0] for both.

```

RTB
RTA  RTB  RTC  RTD
Sum-Net  192.168.30.0  10.2.2.2  431 28  80000001  3
Sum-Net  172.16.102.0  10.2.2.2  487 28  80000001  1

[RTB]display ospf peer

      OSPF Process 1 with Router ID 10.2.2.2
      Neighbors

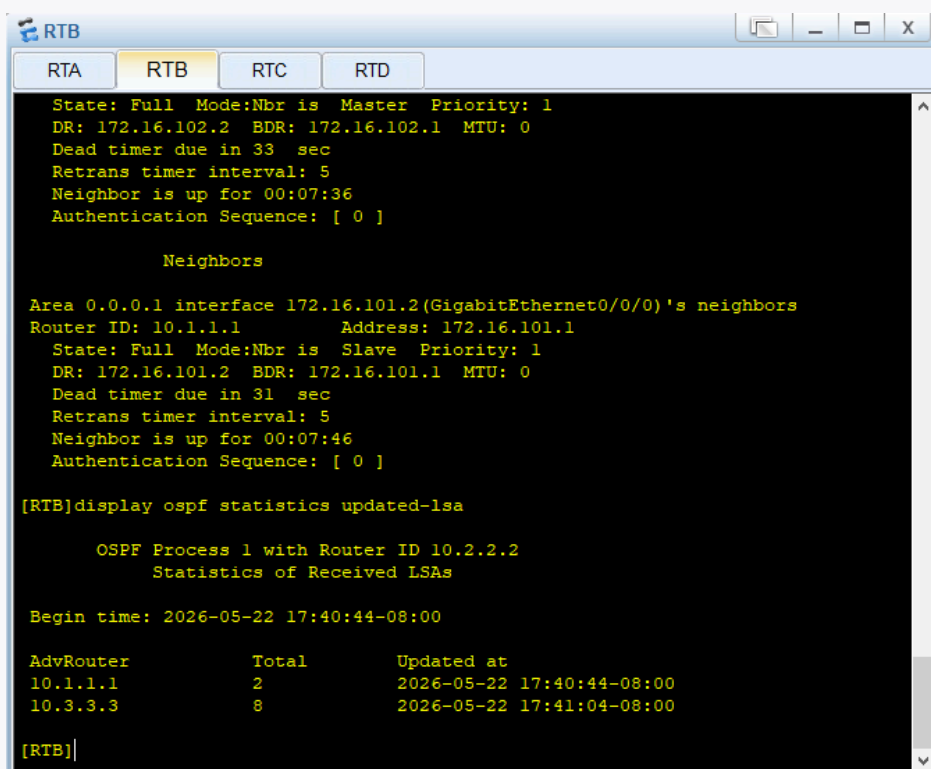
Area 0.0.0.0 interface 172.16.102.1(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.102.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 172.16.102.2  BDR: 172.16.102.1  MTU: 0
  Dead timer due in 33  sec
  Retrans timer interval: 5
  Neighbor is up for 00:07:36
  Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.1 interface 172.16.101.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.1.1.1      Address: 172.16.101.1
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 172.16.101.2  BDR: 172.16.101.1  MTU: 0
  Dead timer due in 31  sec
  Retrans timer interval: 5
  Neighbor is up for 00:07:46
  Authentication Sequence: [ 0 ]

[RTB]

```



The screenshot shows the RTB terminal window with tabs for RTA, RTB, RTC, and RTD. The RTB tab is active. The terminal displays the output of the command `[RTB]display ospf statistics updated-lsa`. It shows OSPF Process 1 with Router ID 10.2.2.2. The statistics of received LSAs are shown for two areas: Area 0.0.0.0 and Area 0.0.0.1. For Area 0.0.0.0, the neighbor is 172.16.102.1 (GigabitEthernet0/0/1) with Router ID 10.3.3.3. For Area 0.0.0.1, the neighbor is 172.16.101.2 (GigabitEthernet0/0/0) with Router ID 10.1.1.1. The state for both is 'Full Mode:Nbr is Master Priority: 1'. The dead timer is 33 seconds for Area 0.0.0.0 and 31 seconds for Area 0.0.0.1. The neighbor is up for 00:07:36 in both areas. The authentication sequence is [0] for both. The begin time is 2026-05-22 17:40:44-08:00. The statistics show that 2 LSAs were received from 10.1.1.1 and 8 LSAs were received from 10.3.3.3.

```

RTB
RTA  RTB  RTC  RTD
State: Full Mode:Nbr is Master  Priority: 1
DR: 172.16.102.2  BDR: 172.16.102.1  MTU: 0
Dead timer due in 33  sec
Retrans timer interval: 5
Neighbor is up for 00:07:36
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.1 interface 172.16.101.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.1.1.1      Address: 172.16.101.1
  State: Full  Mode:Nbr is Slave  Priority: 1
  DR: 172.16.101.2  BDR: 172.16.101.1  MTU: 0
  Dead timer due in 31  sec
  Retrans timer interval: 5
  Neighbor is up for 00:07:46
  Authentication Sequence: [ 0 ]

[RTB]display ospf statistics updated-lsa

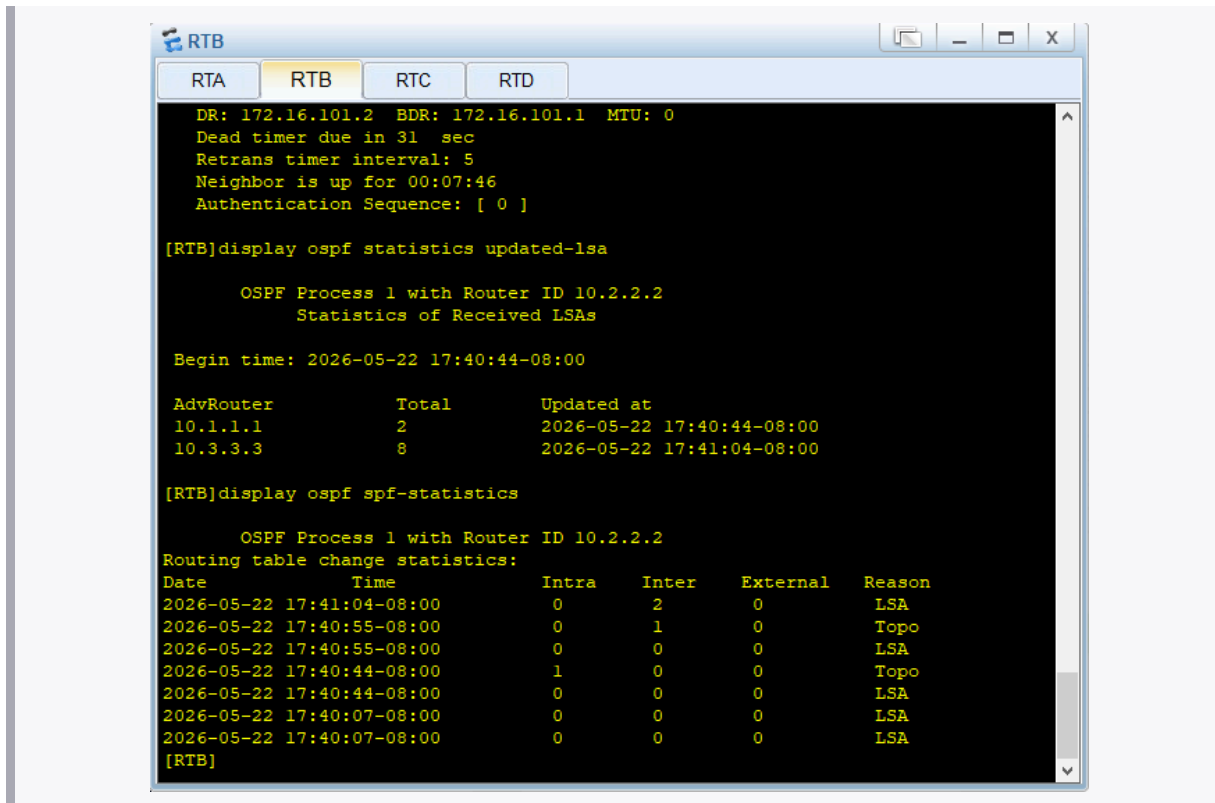
      OSPF Process 1 with Router ID 10.2.2.2
      Statistics of Received LSAs

Begin time: 2026-05-22 17:40:44-08:00

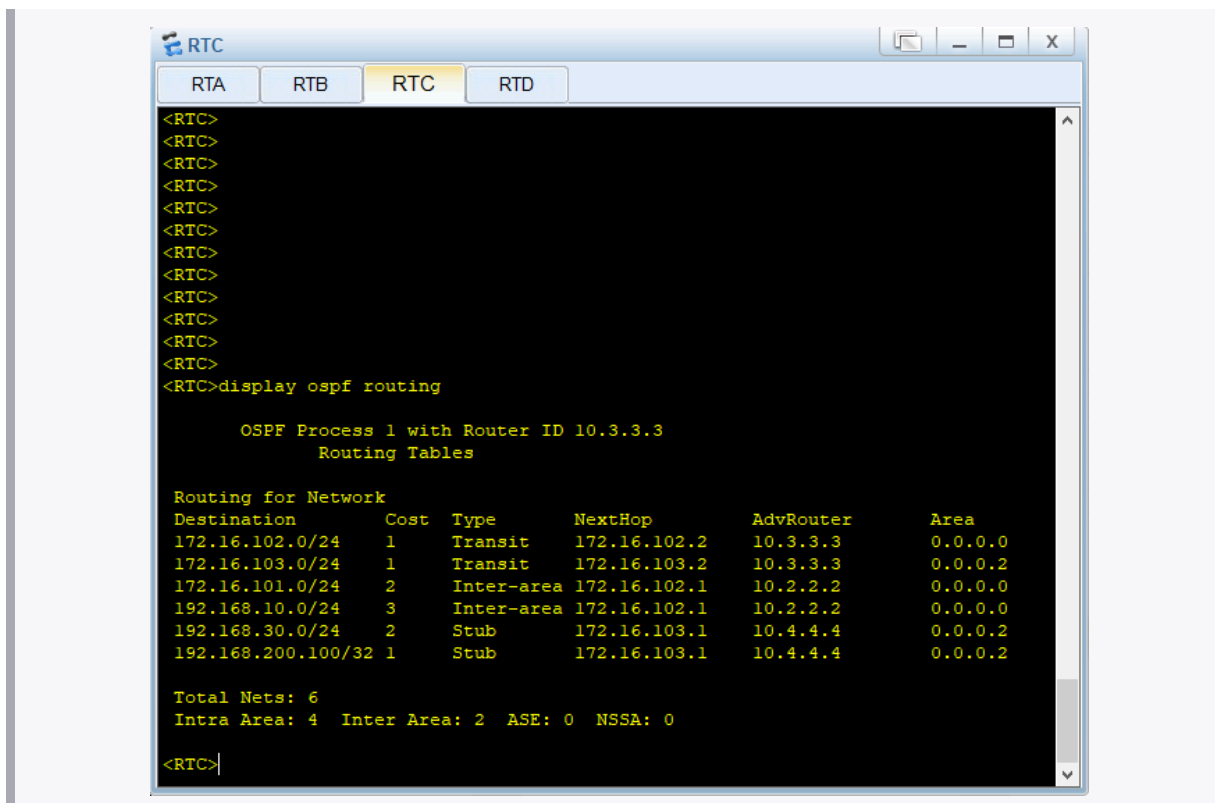
AdvRouter      Total      Updated at
10.1.1.1        2        2026-05-22 17:40:44-08:00
10.3.3.3        8        2026-05-22 17:41:04-08:00

[RTB]

```



Q6-1.4. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTC.



```

RTC
RTA  RTB  RTC  RTD

Total Nets: 6
Intra Area: 4  Inter Area: 2  ASE: 0  NSSA: 0

<RTC>display ospf lsdb

      OSPF Process 1 with Router ID 10.3.3.3
      Link State Database

          Area: 0.0.0.0
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3      10.3.3.3        538  36  800000005      1
Router    10.2.2.2      10.2.2.2        542  36  800000005      1
Network   172.16.102.2    10.3.3.3        538  32  800000002      0
Sum-Net   192.168.200.100 10.3.3.3        532  28  800000001      1
Sum-Net   192.168.10.0   10.2.2.2        552  28  800000001      2
Sum-Net   172.16.103.0   10.3.3.3        581  28  800000001      1
Sum-Net   192.168.30.0   10.3.3.3        532  28  800000001      2
Sum-Net   172.16.101.0   10.2.2.2        589  28  800000001      1

          Area: 0.0.0.2
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3      10.3.3.3        534  36  800000004      1
Router    10.4.4.4      10.4.4.4        525  60  800000007      1
Network   172.16.103.1    10.4.4.4        525  32  800000002      0
Sum-Net   192.168.10.0   10.3.3.3        541  28  800000001      3
Sum-Net   172.16.102.0   10.3.3.3        581  28  800000001      1
Sum-Net   172.16.101.0   10.3.3.3        541  28  800000001      2

<RTC>

```

```

RTC
RTA  RTB  RTC  RTD

Sum-Net   172.16.102.0   10.3.3.3        581  28  800000001      1
Sum-Net   172.16.101.0   10.3.3.3        541  28  800000001      2

<RTC>display ospf peer

      OSPF Process 1 with Router ID 10.3.3.3
      Neighbors

Area 0.0.0.0 interface 172.16.102.2(GigabitEthernet0/0/1)'s neighbors
Router ID: 10.2.2.2      Address: 172.16.102.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 172.16.102.2 BDR: 172.16.102.1 MTU: 0
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00:09:17
Authentication Sequence: [ 0 ]

      Neighbors

Area 0.0.0.2 interface 172.16.103.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.4.4.4      Address: 172.16.103.1
State: Full Mode:Nbr is Master Priority: 1
DR: 172.16.103.1 BDR: 172.16.103.2 MTU: 0
Dead timer due in 31 sec
Retrans timer interval: 5
Neighbor is up for 00:09:10
Authentication Sequence: [ 0 ]

<RTC>

```

```

RTC
RTA  RTB  RTC  RTD

State: Full Mode:Nbr is Slave Priority: 1
DR: 172.16.102.2 BDR: 172.16.102.1 MTU: 0
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00:09:17
Authentication Sequence: [ 0 ]

Neighbors

Area 0.0.0.2 interface 172.16.103.2(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.4.4.4 Address: 172.16.103.1
State: Full Mode:Nbr is Master Priority: 1
DR: 172.16.103.1 BDR: 172.16.103.2 MTU: 0
Dead timer due in 31 sec
Retrans timer interval: 5
Neighbor is up for 00:09:10
Authentication Sequence: [ 0 ]

<RTC>display ospf statistics updated-lsa

OSPF Process 1 with Router ID 10.3.3.3
Statistics of Received LSAs

Begin time: 2026-05-22 17:40:55-08:00

AdvRouter      Total      Updated at
10.2.2.2        4          2026-05-22 17:40:55-08:00
10.4.4.4        5          2026-05-22 17:41:13-08:00

<RTC>

```

```

RTC
RTA  RTB  RTC  RTD

DR: 172.16.103.1 BDR: 172.16.103.2 MTU: 0
Dead timer due in 31 sec
Retrans timer interval: 5
Neighbor is up for 00:09:10
Authentication Sequence: [ 0 ]

<RTC>display ospf statistics updated-lsa

OSPF Process 1 with Router ID 10.3.3.3
Statistics of Received LSAs

Begin time: 2026-05-22 17:40:55-08:00

AdvRouter      Total      Updated at
10.2.2.2        4          2026-05-22 17:40:55-08:00
10.4.4.4        5          2026-05-22 17:41:13-08:00

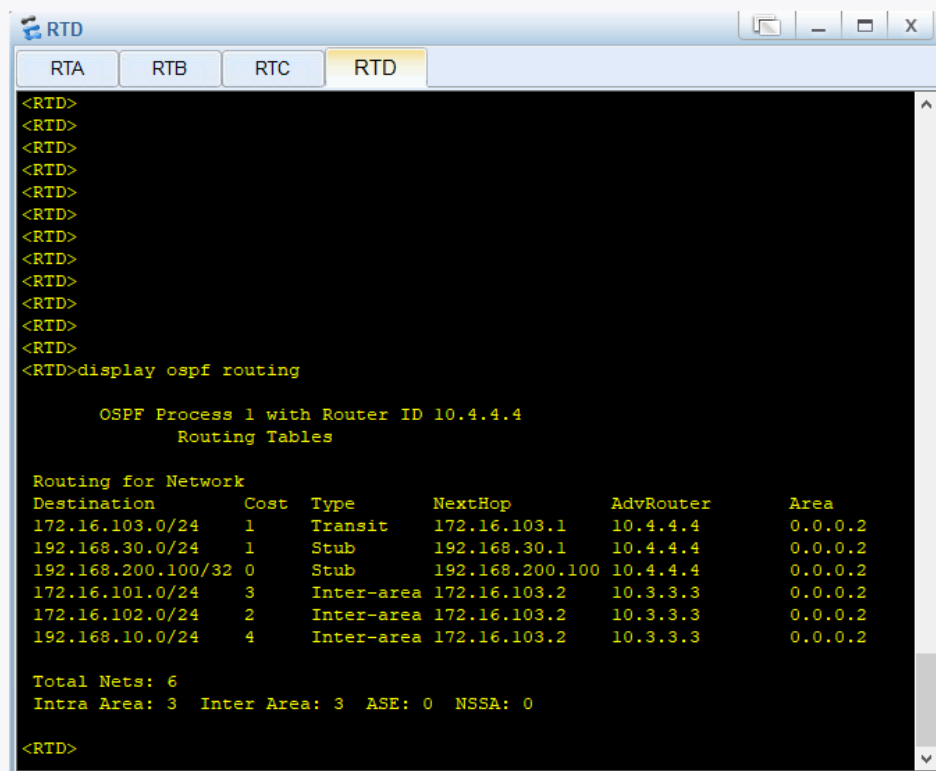
<RTC>display ospf spf-statistics

OSPF Process 1 with Router ID 10.3.3.3
Routing table change statistics:
Date           Time           Intra   Inter   External Reason
2026-05-22 17:41:04-08:00      2       0       0      Topo
2026-05-22 17:41:03-08:00      0       0       0      Topo
2026-05-22 17:41:03-08:00      0       0       0      LSA
2026-05-22 17:40:56-08:00      0       2       0      Topo
2026-05-22 17:40:55-08:00      0       0       0      LSA
2026-05-22 17:40:16-08:00      0       0       0      LSA
2026-05-22 17:40:15-08:00      0       0       0      LSA

<RTC>

```


Q6-1.5. Paste the screenshots of the OSPF routing table, Link-State Database, OSPF neighbor, updates of the LSAs that the LSDB receives and route calculation statistics on the router RTD.



```

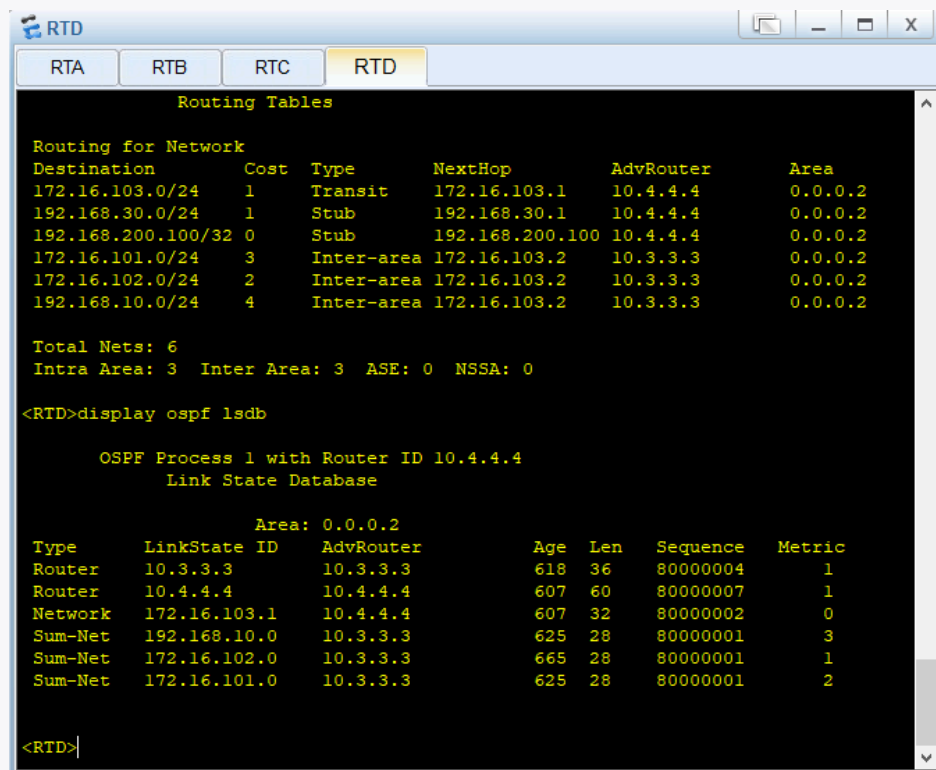
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>display ospf routing

      OSPF Process 1 with Router ID 10.4.4.4
        Routing Tables

Routing for Network
Destination      Cost  Type      NextHop      AdvRouter      Area
172.16.103.0/24  1     Transit   172.16.103.1  10.4.4.4        0.0.0.2
192.168.30.0/24  1     Stub      192.168.30.1  10.4.4.4        0.0.0.2
192.168.200.100/32 0     Stub      192.168.200.100 10.4.4.4        0.0.0.2
172.16.101.0/24   3     Inter-area 172.16.103.2  10.3.3.3        0.0.0.2
172.16.102.0/24   2     Inter-area 172.16.103.2  10.3.3.3        0.0.0.2
192.168.10.0/24   4     Inter-area 172.16.103.2  10.3.3.3        0.0.0.2

Total Nets: 6
Intra Area: 3  Inter Area: 3  ASE: 0  NSSA: 0

<RTD>
  
```



```

<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>
<RTD>display ospf lsdb

      OSPF Process 1 with Router ID 10.4.4.4
        Link State Database

          Area: 0.0.0.2
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3          10.3.3.3        618  36   800000004      1
Router    10.4.4.4          10.4.4.4        607  60   800000007      1
Network   172.16.103.1       10.4.4.4        607  32   800000002      0
Sum-Net   192.168.10.0       10.3.3.3        625  28   800000001      3
Sum-Net   172.16.102.0       10.3.3.3        665  28   800000001      1
Sum-Net   172.16.101.0       10.3.3.3        625  28   800000001      2

<RTD>
  
```

```

RTD
RTA  RTB  RTC  RTD
<RTD>display ospf lsdb

    OSPF Process 1 with Router ID 10.4.4.4
      Link State Database

Area: 0.0.0.2
Type      LinkState ID  AdvRouter      Age  Len  Sequence      Metric
Router    10.3.3.3      10.3.3.3        618  36   800000004      1
Router    10.4.4.4      10.4.4.4        607  60   800000007      1
Network   172.16.103.1      10.4.4.4        607  32   800000002      0
Sum-Net   192.168.10.0      10.3.3.3        625  28   800000001      3
Sum-Net   172.16.102.0      10.3.3.3        665  28   800000001      1
Sum-Net   172.16.101.0      10.3.3.3        625  28   800000001      2

<RTD>display ospf peer

    OSPF Process 1 with Router ID 10.4.4.4
      Neighbors

Area 0.0.0.2 interface 172.16.103.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.103.2
  State: Full Mode:Nbr is Slave Priority: 1
  DR: 172.16.103.1 BDR: 172.16.103.2 MTU: 0
  Dead timer due in 35 sec
  Retrans timer interval: 5
  Neighbor is up for 00:10:31
  Authentication Sequence: [ 0 ]

<RTD>

```

```

RTD
RTA  RTB  RTC  RTD
Sum-Net  192.168.10.0  10.3.3.3      625  28   800000001      3
Sum-Net  172.16.102.0  10.3.3.3      665  28   800000001      1
Sum-Net  172.16.101.0  10.3.3.3      625  28   800000001      2

<RTD>display ospf peer

    OSPF Process 1 with Router ID 10.4.4.4
      Neighbors

Area 0.0.0.2 interface 172.16.103.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.103.2
  State: Full Mode:Nbr is Slave Priority: 1
  DR: 172.16.103.1 BDR: 172.16.103.2 MTU: 0
  Dead timer due in 35 sec
  Retrans timer interval: 5
  Neighbor is up for 00:10:31
  Authentication Sequence: [ 0 ]

<RTD>display ospf statistics updated-lsa

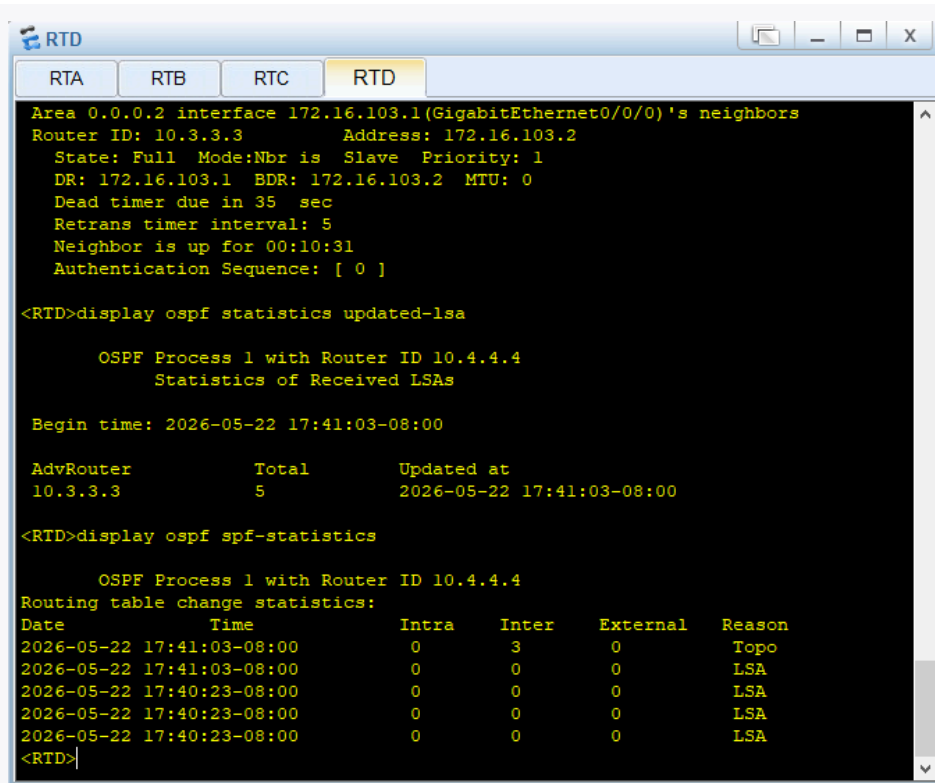
    OSPF Process 1 with Router ID 10.4.4.4
      Statistics of Received LSAs

Begin time: 2026-05-22 17:41:03-08:00

AdvRouter      Total      Updated at
10.3.3.3        5          2026-05-22 17:41:03-08:00

<RTD>

```



```

RTD
RTA  RTB  RTC  RTD
Area 0.0.0.2 interface 172.16.103.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.3.3.3      Address: 172.16.103.2
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 172.16.103.1  BDR: 172.16.103.2  MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:10:31
Authentication Sequence: [ 0 ]

<RTD>display ospf statistics updated-lsa

      OSPF Process 1 with Router ID 10.4.4.4
      Statistics of Received LSAs

Begin time: 2026-05-22 17:41:03-08:00

AdvRouter      Total      Updated at
10.3.3.3        5        2026-05-22 17:41:03-08:00

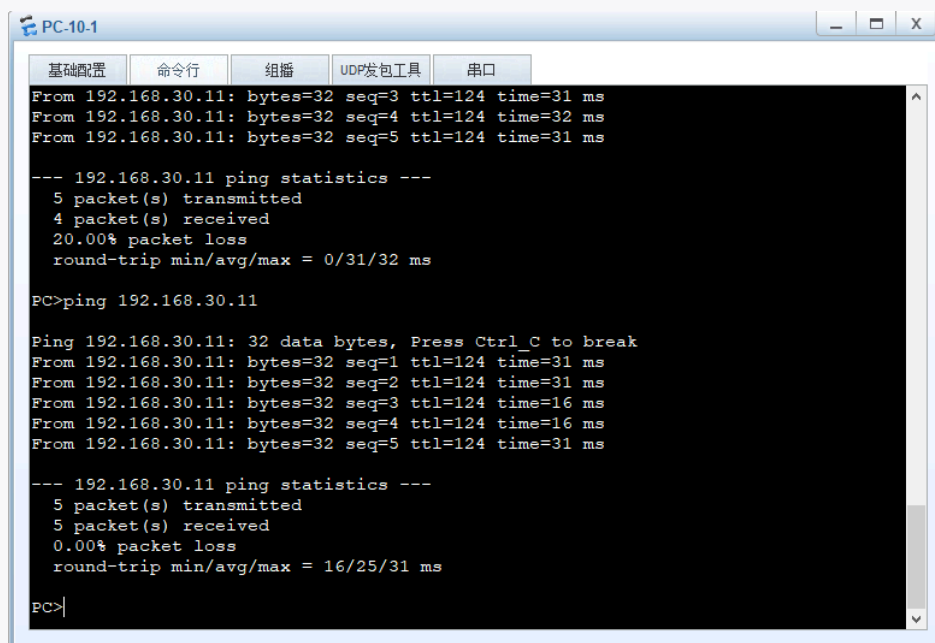
<RTD>display ospf spf-statistics

      OSPF Process 1 with Router ID 10.4.4.4
Routing table change statistics:
Date           Time           Intra   Inter   External  Reason
2026-05-22 17:41:03-08:00      0       3       0       Topo
2026-05-22 17:41:03-08:00      0       0       0       LSA
2026-05-22 17:40:23-08:00      0       0       0       LSA
2026-05-22 17:40:23-08:00      0       0       0       LSA
2026-05-22 17:40:23-08:00      0       0       0       LSA
<RTD>

```

Q6-1.6. Can PC-10-1 and PC-30-1 ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



```

PC-10-1
基础配置  命令行  组播  UDP发包工具  串口
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=32 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/31/32 ms

PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=2 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=16 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=16 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

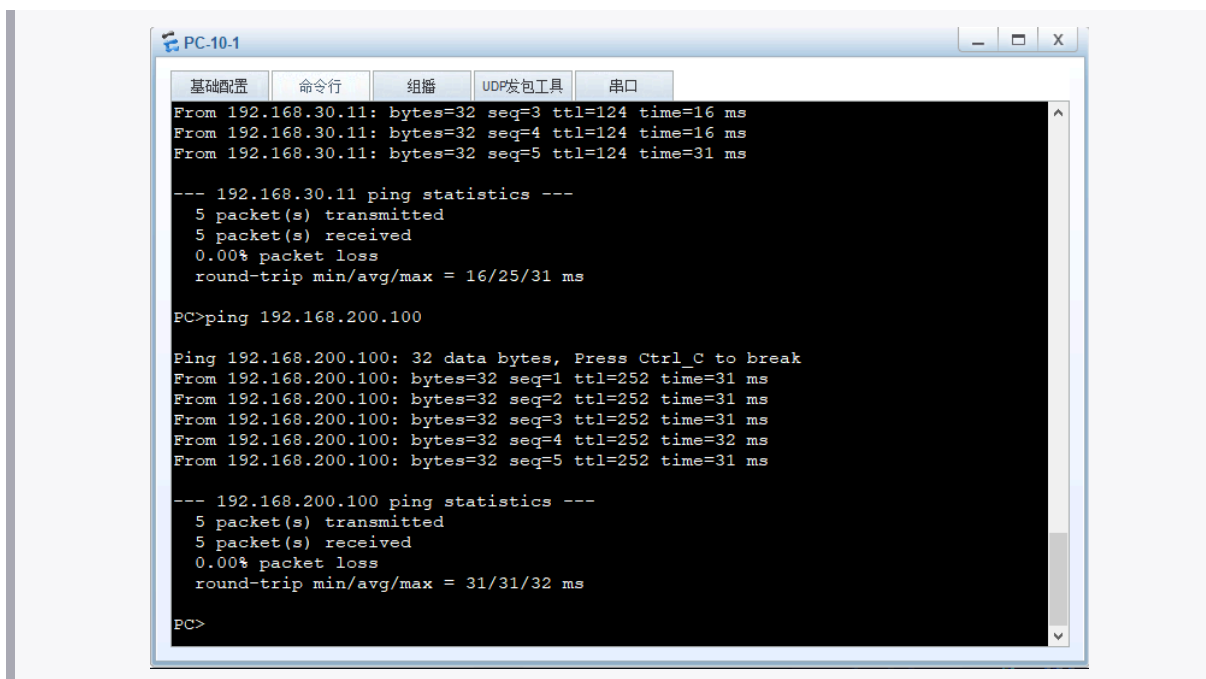
--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 16/25/31 ms

PC>

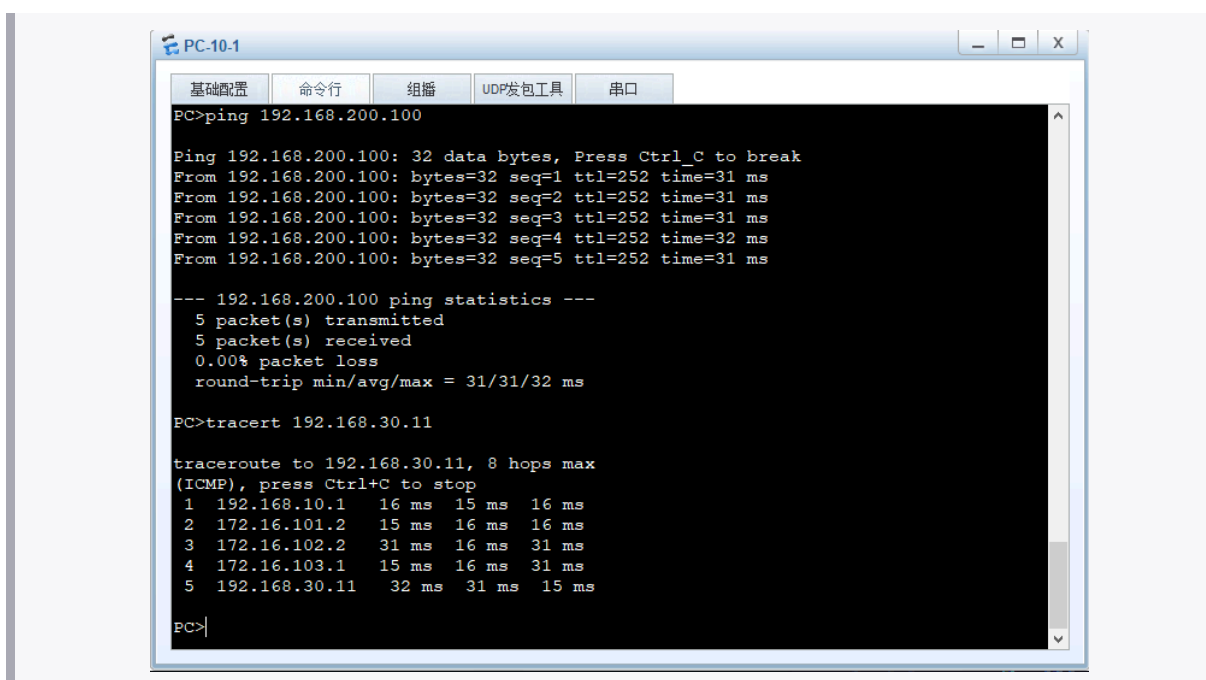
```

Q6-1.7. Can PC-10-1 and loopback 0 on RTD ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



Q6-1.8. Please paste the screenshot of the result of command “tracert 192.168.30.11” issued from PC-10-1.



Q6-1.9. Just after configuring OSPF on RTA and RTB, what OSPF packets are exchanged between RTA and RTB? Please explain with the captured OSPF communications.

在刚配置 OSPF 时，RTA 与 RTB 之间互相交换了 5 种 OSPF 报文：

- 发送 **Hello** 报文，用于发现和建立邻居关系。
- 发送 **Database Description (DD)** 报文，用于交换链路状态数据库 (LSDB) 的摘要信息。
- 发送 **Link State Request (LSR)** 报文，用于向对方请求自身缺失的链路状态通告 (LSA)。

- 发送 **Link State Update (LSU)** 报文，用于发送完整的 LSA 详细信息给对方。
- 发送 **Link State Acknowledgment (LSAck)** 报文，用于对收到的 LSU 进行确认。

No.	Time	Source	Destination	Protocol	Length	Info
213	672.656000	172.16.101.2	224.0.0.5	OSPF	78	Hello Packet
216	674.453000	172.16.101.2	255.255.255.255	RIPv1	66	Response
217	678.156000	172.16.101.1	224.0.0.5	OSPF	82	Hello Packet
218	678.172000	172.16.101.2	172.16.101.1	OSPF	66	DB Description
219	678.187000	172.16.101.1	172.16.101.2	OSPF	66	DB Description
220	678.197000	172.16.101.1	172.16.101.2	OSPF	206	DB Description
221	678.187000	172.16.101.1	224.0.0.5	OSPF	110	LS Update
222	678.203000	172.16.101.2	172.16.101.1	OSPF	142	LS Request
223	678.203000	172.16.101.2	172.16.101.1	OSPF	166	DB Description
224	678.219000	172.16.101.1	172.16.101.2	OSPF	290	LS Update
225	678.219000	172.16.101.1	172.16.101.2	OSPF	66	DB Description
226	678.219000	172.16.101.2	224.0.0.5	OSPF	242	LS Update
227	678.219000	172.16.101.1	224.0.0.5	OSPF	142	LS Update
228	678.422000	172.16.101.1	224.0.0.5	OSPF	178	LS Acknowledge
229	678.672000	172.16.101.2	172.16.101.1	OSPF	78	LS Acknowledge
230	678.672000	172.16.101.2	224.0.0.5	OSPF	98	LS Acknowledge
231	679.234000	172.16.101.2	224.0.0.5	OSPF	98	LS Update
232	679.422000	172.16.101.1	224.0.0.5	OSPF	78	LS Acknowledge
233	682.073000	172.16.101.2	255.255.255.255	RIPv1	126	Response
234	683.234000	172.16.101.2	224.0.0.5	OSPF	12	Hello Packet
235	683.250000	172.16.101.1	224.0.0.5	OSPF	142	LS Update

Q6-1.10. With the completion of the OSPF configurations on each router, how does RTA know that the network topology has changed? Please explain with the captured OSPF communications.

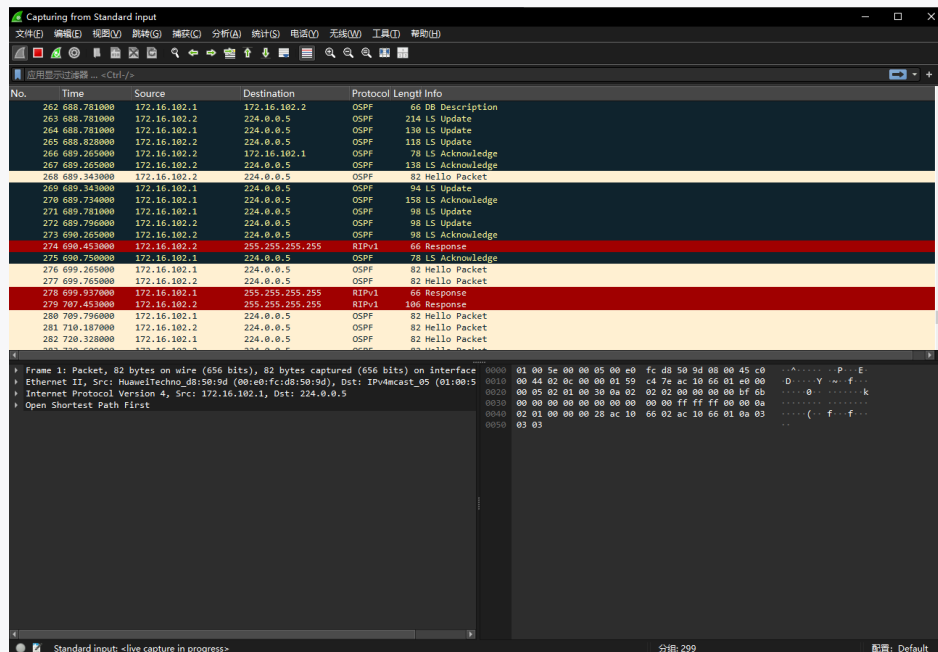
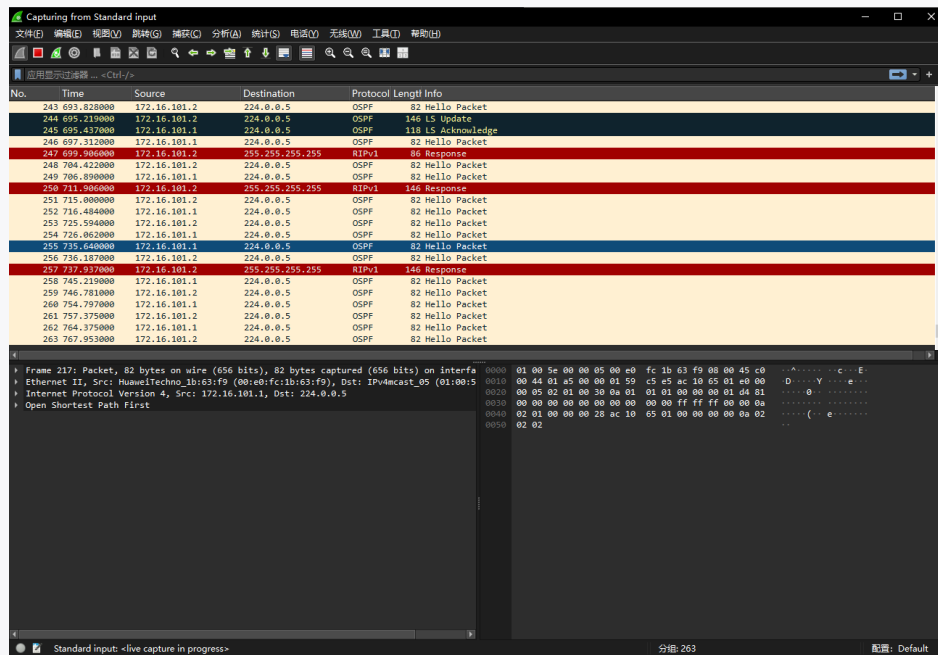
OSPF 协议基于触发更新 (Triggered Updates) 机制。当网络拓扑发生改变（例如有新路由器加入或路由宣告完成）时，感知到变化的路由器（如 RTB）会立即向目标组播地址 224.0.0.5 发送装载了最新链路状态通告 (LSA) 的 **LSU** 报文进行泛洪。

RTA 接收到该 LSU 报文后，会更新自身的链路状态数据库 (LSDB)，并重新运行 **SPF (Dijkstra)** 算法，计算出新的最短路径并更新路由表。

Q6-1.11. What OSPF packets are sent and received from the interface GE 0/0/0 and GE 0/0/1 on RTB?

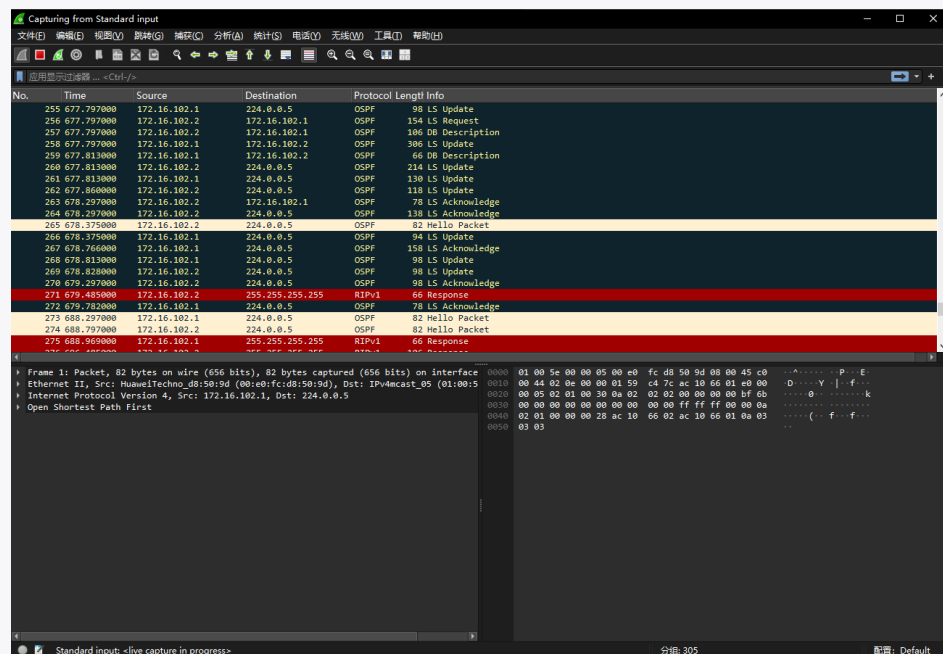
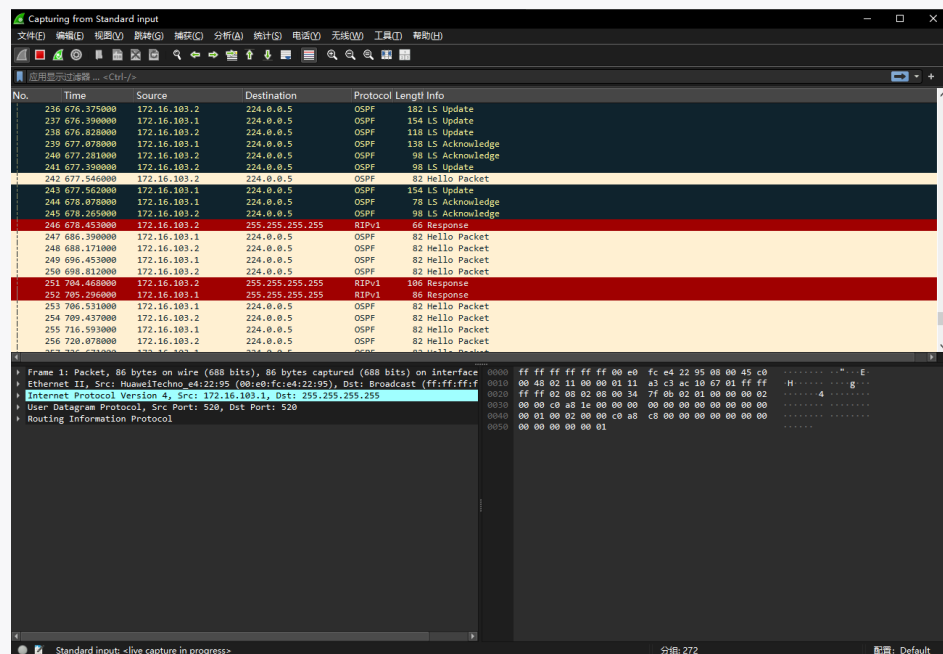
Paste the screenshot of the captured OSPF packets.

RTB 的 GE 0/0/0 和 GE 0/0/1 接口发送和接收了 Hello, DB Description (DD), LS Request (LSR), LS Update (LSU) 以及 LS Acknowledge (LSAck) 报文，绝大多数报文的目标 IP 地址为 OSPF 组播地址 224.0.0.5。



Q6-1.12. What OSPF packets are sent and received from the interface GE 0/0/0 and GE 0/0/1 on RTC? Paste the screenshot of the captured OSPF packets.

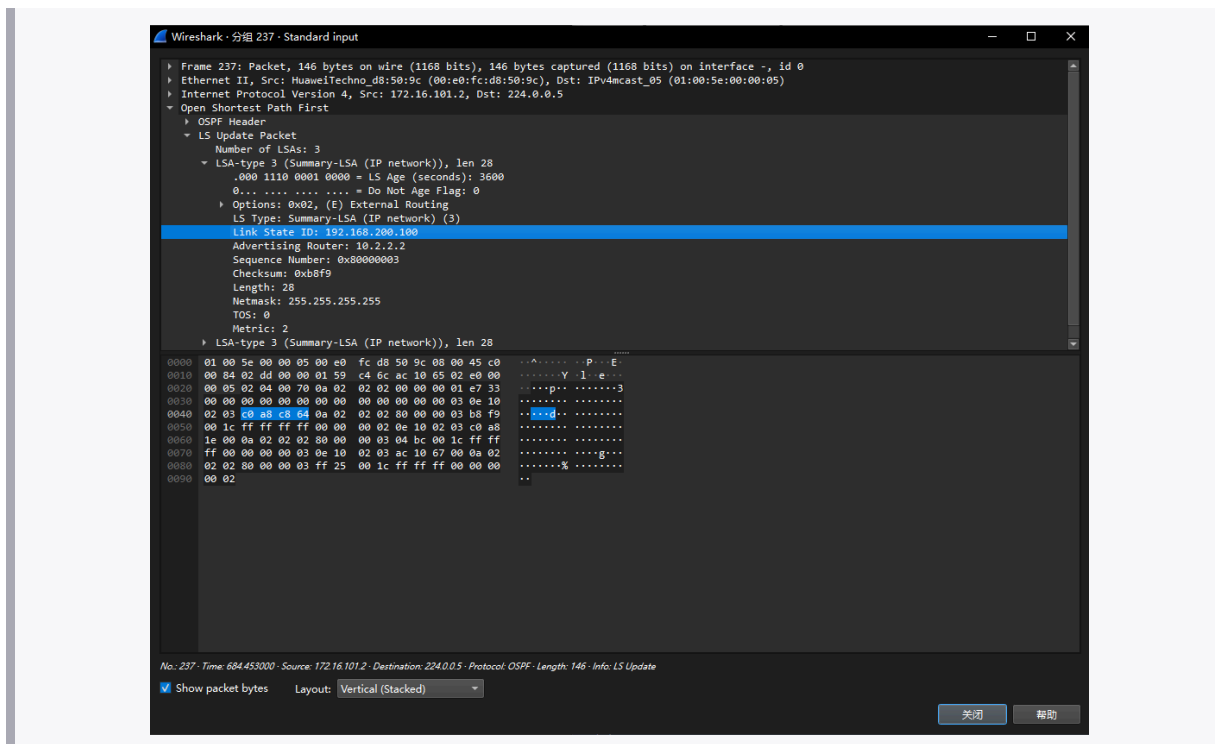
与 RTB 类似, RTC 的 GE 0/0/0 和 GE 0/0/1 接口同样发送和接收了全部 5 种 OSPF 报文 (Hello, DD, LSR, LSU, LSAck), 目标地址主要是组播地址 224.0.0.5。



Q6-1.13. In what type of packet are the OSPF Link State Advertisements sent from the interface GE 0/0/0 on RTB transmitted? What are their LSA types? Paste the screenshot of the expanded information of the captured LSAs.

OSPF 的链路状态通告 (LSA) 是被封装在 **LS Update (LSU)** 报文中进行传输的。

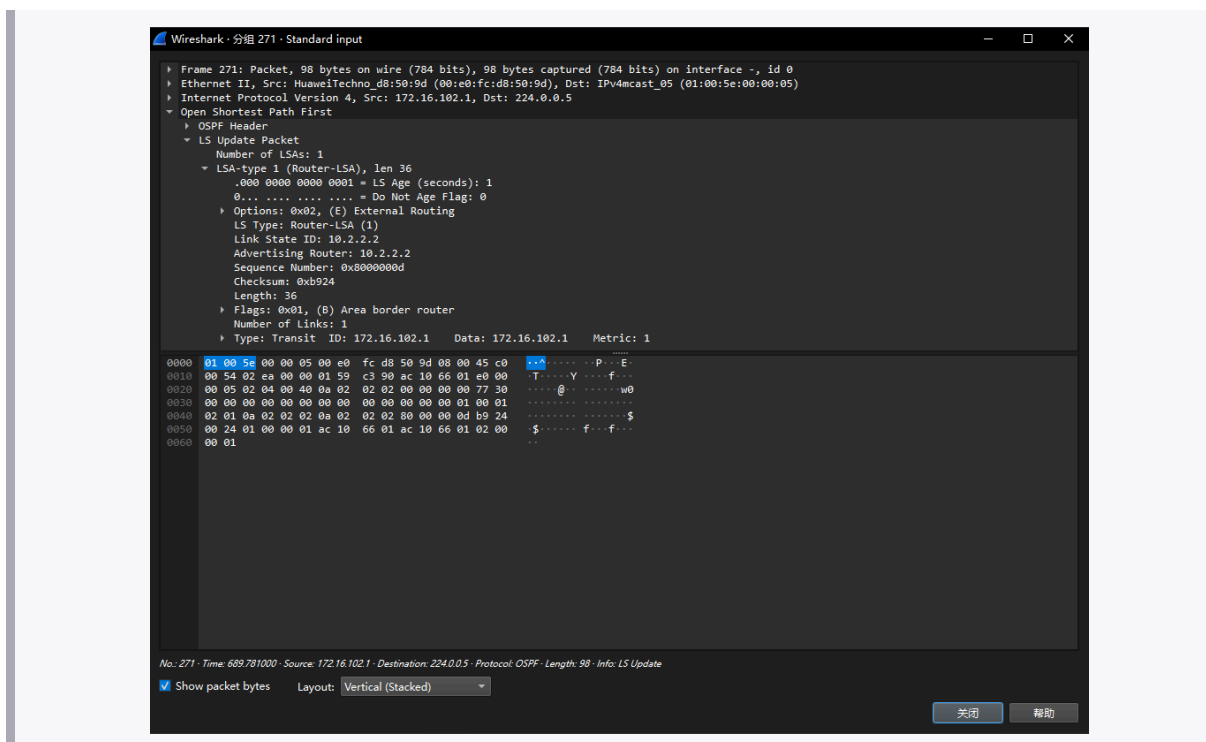
从截图中可以看出，在 RTB 的 GE 0/0/0 接口（连接 Area 1，源地址为 172.16.101.2）发送的报文中，包含了 **Type 3 (Summary-LSA)** 类型的 LSA。这是因为 RTB 作为区域边界路由器 (ABR)，将其学习到的其他区域的路由信息（如图中展开的属于 Area 2 的 192.168.200.100）转换为 Type 3 LSA，并通过该接口注入到 Area 1 中供 RTA 学习。



Q6-1.14. In what type of packet are the OSPF Link State Advertisements sent from the interface GE 0/0/1 on RTB transmitted? What are their LSA types? Paste the screenshot of the expanded information of the captured LSAs.

LSA 同样是被封装在 **LS Update (LSU)** 报文中传输的。

从截图中可以看出，在 RTB 的 GE 0/0/1 接口（连接骨干区域 Area 0，源地址为 172.16.102.1）发送的报文中，包含了 **Type 1 (Router-LSA)** 类型的 LSA。RTB 作为 Area 0 的一员，需要发送 Type 1 LSA 来描述自身在骨干区域的链路状态（截图中展开的 Flags: (B) Area border router 标识清晰地证明了它是 ABR）。



Q6-1.15. How are the Link State Advertisements in area 1 transmitted to area 2 through area 0? Please explain with the captured OSPF communications.

- 首先，Area 1 内的 RTA 会生成 **Type 1 Router-LSA**，用于描述其直连网络并在 Area 1 内部进行泛洪。
- 其次，RTB 作为连接 Area 1 和 Area 0 的区域边界路由器 (ABR)，在收到该 Type 1 LSA 后，会将其转换为 **Type 3 Summary-LSA**，并将该路由汇总信息注入到骨干区域 Area 0 中传输。
- 最后，RTC 作为另一端的 ABR，在 Area 0 中接收到该 Type 3 LSA 后，会重新生成新的 Type 3 LSA 并将其泛洪到 Area 2 中。最终 RTD 收到该通告，从而学习到了通往 Area 1 的路由。

Q6-1.16. How are the Link State Advertisements in area 2 transmitted to area 1 through area 0? Please explain with the captured OSPF communications.

传递机制与上一题的逻辑完全相同，方向相反：

- 首先，Area 2 内的 RTD 会生成 **Type 1 Router-LSA** 来描述其网络（包含 192.168.30.0/24 以及 Loopback 0 接口的网络），并在 Area 2 内泛洪。
- 其次，RTC 作为 ABR，在收到该通告后将其打包转换为 **Type 3 Summary-LSA**，并将其泛洪至骨干区域 Area 0。
- 最后，RTB 从 Area 0 中接收到该 Type 3 LSA 后，生成新的 Type 3 LSA 并泛洪进 Area 1 中，从而使得 RTA 最终学习到通往 Area 2 的跨区域路由。